

Cables and Connectors

Personal Computers

PC components are divided into user-handled peripheral devices and internal components housed in the system case, which could be damaged or dangerous if exposed. Peripherals include input devices (keyboard, mouse, microphone, camera), output devices (reference display/monitor, speakers), and external storage. The case or chassis, typically a vertical tower, contains the internal components, such as the motherboard, CPU, memory modules, adapter cards, fixed disks, and power supply unit.

Note:

PCs can also be purchased as all-in-one units. All-in-one means that the internal components are contained within a case that is also a monitor.

To perform PC maintenance, you must understand how to open a desktop computer's case.

- A tower case has a side cover that slides off, secured by screws or clips, and may have anti-tamper mechanisms. Always refer to system documentation and follow recommended steps.
- The front panel provides access to removable media drives, a power switch, and LEDs. It can be removed but may require removing the side panel first to access its screws or clips.

The front of a typical PC case

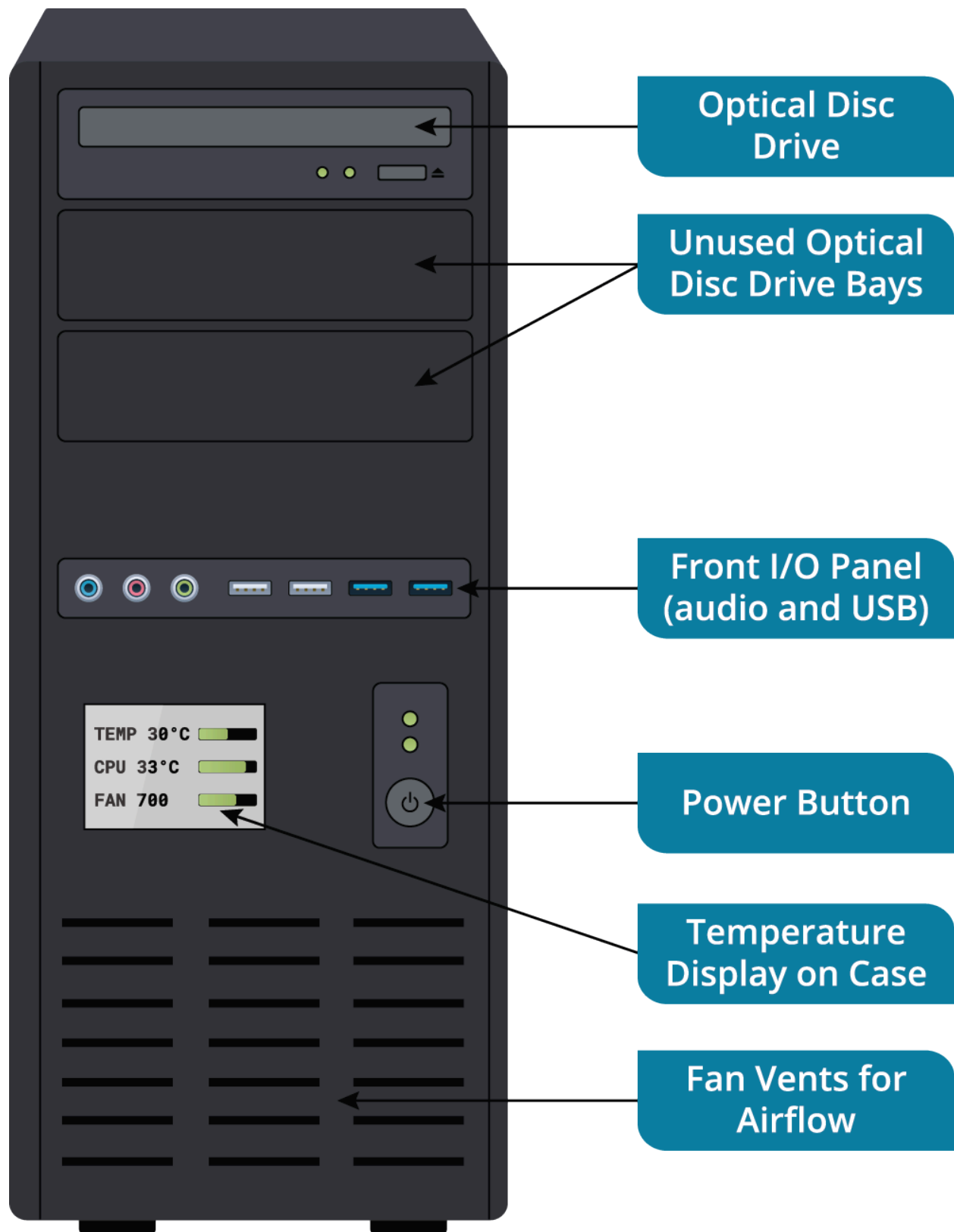


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▼ Description

Optical disc drive positioned near the top of the tower. Unused optical disc drive bays are empty slots below the optical disc drive. Front input or output panel for audio and U S B contains ports for audio and U S B devices located centrally on the front. Power button is a round button. Temperature display on case is a digital panel showing the system's temperature, located beneath the input output panel. Fan vents for airflow are the openings at the bottom of the case.

The rear panel provides access to the power supply unit (PSU) sockets, which include an integral fan exhaust. Ensure the fan is unobstructed to maintain proper cooling. There may also be an additional case fan.

The rear panel of a typical PC case

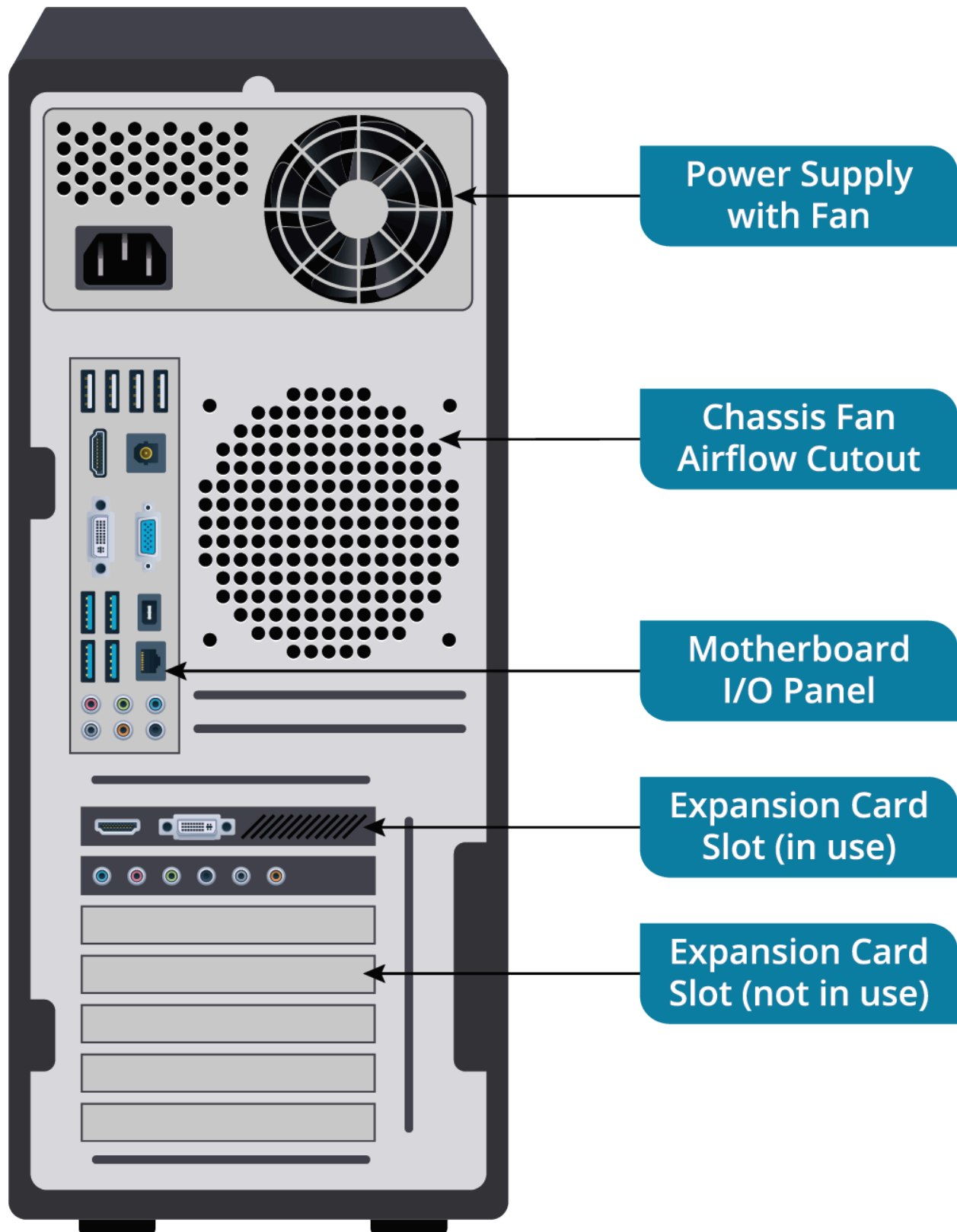


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▼ Description

Power supply with fan is located at the top. Chassis fan airflow cutout is a circular vent below the power supply. Motherboard input output panel is a section containing various ports. Expansion card slot in use and expansion card slot not in use are located below the motherboard input output panel and chassis fan airflow cutout.

Below the PSU, a cutout aligns with the motherboard's input/output (I/O) ports for peripheral connections, covered by an I/O shield to keep out dust. At the bottom of the rear panel, cutout slots align with adapter card slots, covered by either an adapter card or a blanking plate. These shields and plates prevent gaps in the case, which can cause:

- Dust entry, leading to component overheating.
- Increased exposure to electrostatic discharge (ESD), which can damage chips. The I/O shield's pins connect external metal parts to the metal case and PSU, providing a safe path for ESD to ground.
- Increased exposure to electromagnetic interference (EMI). EMI is energy from magnetic and electrical sources, such as motors or other electronic devices, which can cause temporary or permanent faults. The case absorbs EMI, but gaps reduce this protection.

Peripheral Devices

An input/output (I/O) port connects devices to the PC via peripheral cables. Some ports are specific, like a graphics port for a monitor, while others support various devices. External ports are located at the rear or front of the PC through case cutouts and can be on the motherboard or an expansion card.

I/O ports on a motherboard

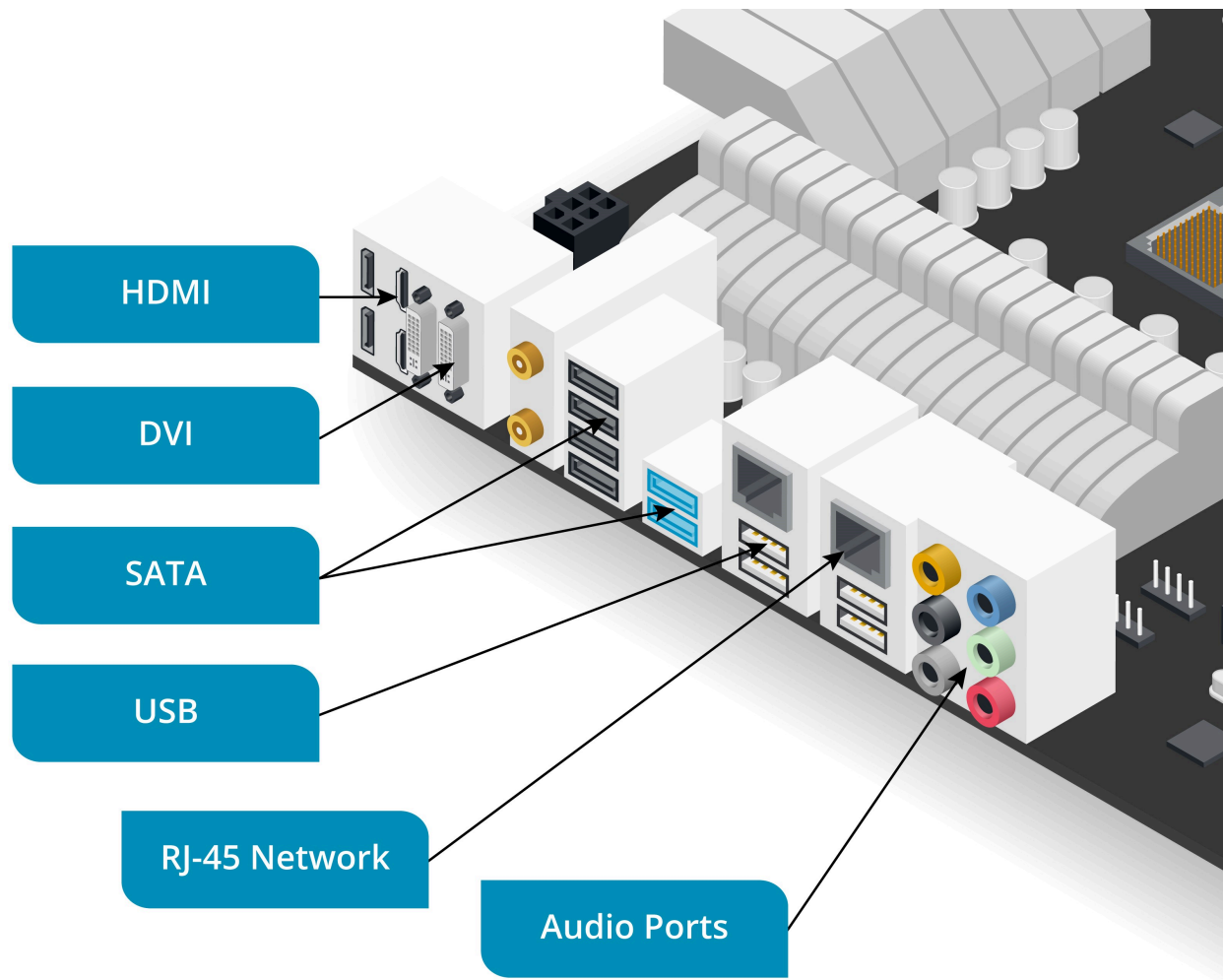


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▼ Description

The ports are as follows: H D M I, D V I, S A T A, U S B, R J-45 Network, and audio ports.

Interfaces, Ports, and Connectors

A hardware port is the external connection point for a bus interface, allowing data transfer to and from devices. The connector is the part of a peripheral cable that fits into a matching port. Each bus interface may use multiple connector types. Most connectors and ports now use edge contacts and are either asymmetric or keyed to prevent incorrect insertion or are reversible.

Peripheral cable for the Universal Serial Bus (USB) interface

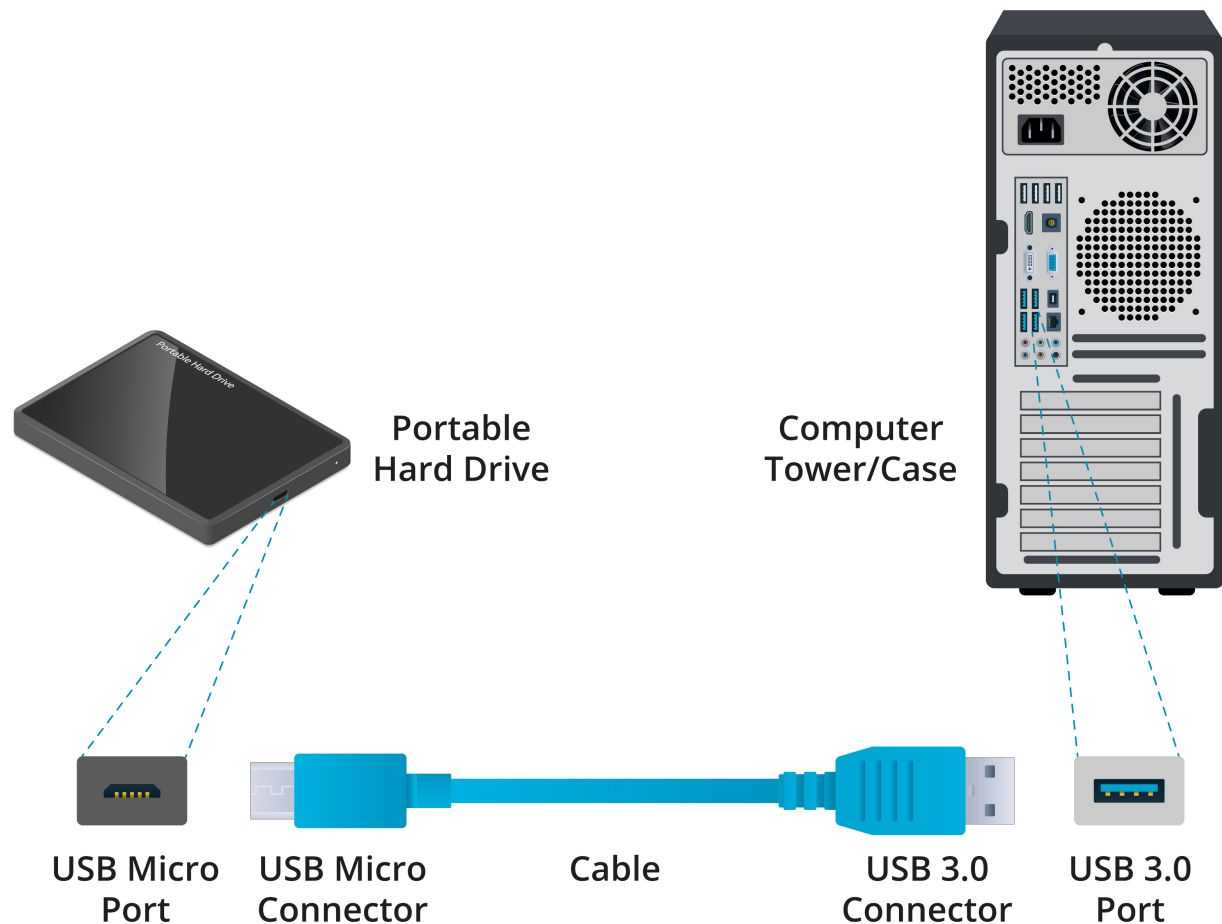


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▼ Description

A portable hard drive has a U S B micro port labeled. A U S B cable is shown with two ends: a U S B micro connector on the left and a U S B 3.0 Connector on the right. A back view of the computer tower has a U S B 3.0 Port.

Binary Data Storage and Transfer units

When comparing bus interfaces, use appropriate units. Computers process binary data, where each digit or bit (b) is a 0 or 1. Storage is measured in bytes (B), which

are eight bits. A lowercase "b" refers to a bit, while an uppercase "B" means a byte.

Transfer rates are expressed in units per second of the following multiples of bits and bytes:

- 1000—Kilobits (Kb/s or Kbps) and kilobytes (KB/s and KBps).
- 1000×1000—Megabits (Mb/s) or megabytes (MB/s).
- 1000×1000×1000—Gigabits (Gb/s) and gigabytes (GB/s).

Universal Serial Bus Cables

The Universal Serial Bus (USB) is the standard for connecting most peripheral devices to a computer. USB devices are categorized into classes like human interface (keyboards and mice), mass storage (disk drives), printers, and audio devices. A USB is managed by a host controller, which supports multiple ports on the same bus. Each controller can theoretically support up to 127 devices, but to overcome bandwidth limitations, most PC motherboards have multiple USB controllers, each with three or four ports.

USB port symbol



USB Connector Types

USB 2 connector form factors include:

- Type A: Flat rectangular connector for host and some peripheral devices, inserted with the USB symbol facing up.
- Type B: Square connector with a beveled top for large devices like printers.

- Type B Mini: Smaller connector for smaller peripheral devices like early digital cameras, but now rarely used.
- Type B Micro: Flatter connector for smaller devices like smartphones and tablets.

USB 2 ports and connectors



Image © 123RF.com

▼ Description

The ports and the connectors are as follows. Type A: A flat rectangular connector with a matching rectangular port. Type B: A square connector with a slightly beveled top and a matching port. Type B Mini: A smaller trapezoidal connector with a matching port. Type B Micro: An elongated trapezoidal connector with a matching port.

A USB cable can have Type A to Type A connectors or convert between types (e.g., Type A to Type B or Type A to Micro Type B).

USB 3 introduces new versions of Type A, Type B, and Type B Micro connectors with additional signaling pins and wires, often distinguished by a blue connector tab or housing. USB 3.0 Type A connections are physically compatible with USB 1.1 and 2.0, but Type B and Type B Micro connections are not. For example, you can plug a USB 2 Type A cable into a USB 3 Type A port, but not a USB 3 Type B cable into a USB 2 Type B port.

USB 3 connectors and ports (from left to right): Type A, Type B, Micro Type B, Type C



Image © 123RF.com

▼ Description

The ports and the connectors are as follows. Type A: A flat rectangular connector with a matching rectangular port. Type B: A square connector with a slightly beveled top and a matching port. Type B Micro: A smaller connector with a dual-layer design with a matching port. Type C: A connector with rounded edges and a thunderbolt symbol on it with a matching port.

USB 3.1 introduces the USB-C connector, a compact, reversible, and durable design aimed at providing a consistent hardware interface. USB-C connectors can be used at both ends of a cable or paired with converter cables to connect to USB Type A or Type B ports.

Cable Length

The maximum cable length is 3 meters for LowSpeed devices and 5 meters for FullSpeed and HighSpeed devices. While vendors may offer longer cables, performance may degrade beyond these lengths. For SuperSpeed-capable cables, a recommended length is up to 3 meters, though there is no official maximum.

Power

In addition to supplying a data signal, the USB bus can supply power to connected devices. Most USB Type A and Type C ports can charge device batteries.

Note:

Basic USB ports can provide up to 4.5 watts, depending on the version. Power Delivery (PD) ports can supply up to 100 watts with suitable connectors and cables.

USB Standards

The evolution of the USB standard has brought significant improvements in speed, power delivery, and functionality:

1. USB 1.0 (1996): Introduced with speeds of 1.5 Mbps (Low Speed) and 12 Mbps (Full Speed), using Type-A and Type-B connectors.
2. USB 1.1 (1998): Enhanced USB 1.0 with the same speeds but improved compatibility and reliability.
3. USB 2.0 (2000): Increased speed to 480 Mbps (High Speed) and introduced Mini-A/B and Micro-A/B connectors.
4. USB 3.0 (2008): Introduced SuperSpeed at 5 Gbps and new connectors like Micro-B, later rebranded as USB 3.1 Gen 1.
5. USB 3.1 (2013): Brought SuperSpeed+ at 10 Gbps and the versatile Type-C connector, rebranded as USB 3.1 Gen 2.
6. USB 3.2 (2017): Offered Gen 1 (5 Gbps), Gen 2 (10 Gbps), and Gen 2×2 (20 Gbps) speeds, using Type-A, Type-B, and Type-C connectors.
7. USB4 (2019): Achieved speeds up to 40 Gbps, unified with Thunderbolt 3, exclusively using the Type-C connector, and supported advanced features like multiple data and display protocols.

Note:

SuperSpeed branding has been deprecated in favor of simpler bitrate labels: USB 5Gbps, USB 10Gbps, USB 20Gbps, and USB 40Gbps.

Display Types

Display technologies have evolved significantly, offering various types to meet different needs. Liquid Crystal Displays (LCDs) are the most common, with three main subtypes:

- **In-Plane Switching (IPS):** A type of liquid crystal display technology where the liquid crystals are aligned parallel to the screen, allowing for consistent light transmission and color reproduction.
- **Twisted Nematic (TN):** A type of liquid crystal display technology where the liquid crystals twist 90 degrees between the electrodes to control light passage and create images on the screen.
- **Vertical Alignment (VA):** A type of liquid crystal display technology where the liquid crystals are aligned vertically to the glass substrates and tilt when voltage is applied to control light passage and create images.

In addition to LCDs, display technologies include OLED and Mini-LED options.

An **Organic Light-Emitting Diode (OLED)** display is a type of display technology where each pixel emits its own light through organic compounds when an electric current is applied. A **Mini-LED** display is a type of display technology that uses thousands of tiny LEDs for backlighting, allowing for more precise control of brightness and contrast in the image.

When comparing display technologies, it is essential to consider the pros and cons of IPS-LCD, TN-LCD, VA-LCD, OLED, and Mini-LED displays to determine which option best meets specific needs for color accuracy, viewing angles, contrast, and overall performance.

- In-Plane Switching (IPS) LCD:
 - Pros: Superior color accuracy and wide viewing angles, making them ideal for graphic design and professional use.
 - Cons: Typically, slower response times compared to TN panels, which may affect fast-paced gaming.
- Twisted Nematic (TN) LCD:
 - Pros: Faster response times and higher refresh rates, which are preferred for gaming.

- Cons: Poorer color reproduction and limited viewing angles.
- Vertical Alignment (VA) LCD:
 - Pros: Better color accuracy and viewing angles than TN.
 - Cons: Slower response times than TN but better than IPS in some cases. VA panels can exhibit color shifting when viewed from extreme angles.
- Organic Light-Emitting Diode (OLED):
 - Pros: Excellent color accuracy, contrast ratios, energy efficiency, and true blacks due to self-emissive pixels.
 - Cons: Potential for burn-in, where static images can leave permanent marks if displayed for long periods.
- Mini-LED Technology:
 - Pros: Mini-LED technology offers improved brightness, better contrast and black levels, enhanced color accuracy, and reduced halo effect, allowing for thin and lightweight display designs suitable for a variety of applications.
 - Cons: Mini-LED displays are more expensive, can consume more power than OLEDs, and while they reduce blooming, they do not achieve the true blacks of OLED technology and add complexity to the manufacturing process.

Note:

Mini-LED is an enhancement for LCDs, not a type of OLED.

Display Components

Understanding display components and attributes is essential for evaluating the performance and suitability of various screens. Key elements include touch screens and digitizers for interactivity, inverters for older LCD backlighting, and critical attributes like pixel density, refresh rates, and screen resolution that determine image quality and clarity.

1. Touch Screens and Digitizers:

- Integral for interactive displays, allowing direct interaction through touch or stylus. This is common in smartphones, tablets, and some laptops.

2. Inverter:

- Crucial in older LCDs for converting DC power to AC for the backlight. Modern displays with LED backlighting typically do not require an inverter.

3. Pixel Density, Refresh Rates, and Screen Resolution:

- Pixel Density: Higher density results in sharper images.
- Refresh Rates: Higher rates (measured in Hz) provide smoother motion, which is important for gaming and video playback.
- Screen Resolution: Defines clarity and detail of the display. Common resolutions include:
 - Full HD (1920×1080)
 - Quad HD (2560×1440)
 - 4K (3840×2160)
 - Newer Resolutions: 5K (5120×2880) and 8K (7680×4320) are emerging, offering even higher clarity and detail.
- Color Gamut: Refers to the range of colors a display can produce based on the RGB color model. A wider color gamut allows for more vibrant and accurate color reproduction, enhancing the visual experience for activities such as photo editing, video production, and gaming. Common color gamuts include sRGB, Adobe RGB, and DCI-P3, with DCI-P3 offering a broader range of colors compared to sRGB. Additionally, color depth plays a crucial role, with 24-bit color (often referred to as true color) supporting approximately 16.7 million colors, while 32-bit color includes an alpha channel for transparency, providing even more detailed color representation and smoother gradients.

Video Cable Bandwidth

Video cable bandwidth is determined by two main factors:

- Resolution: Measured in pixels (e.g., 1920×1080 for Full HD, 3840×2160 for 4K).
- Refresh Rate: Measured in hertz (Hz) or frames per second (fps).

For example, uncompressed HD video at 60 fps requires 4.5 Gbps, while 4K at 60 fps requires 8.91 Gbps.

Note:

Frame rate (fps) describes the video source, while hertz (Hz) indicates the display's refresh rate. To avoid artifacts like ghosting and tearing, the refresh rate should match or be divisible by the frame rate. For example, a 60 fps video will play smoothly on a 120 Hz display.

Note:

An LCD/TFT is often called a flat-panel or LED display, named after its backlight technology (older models use fluorescent tubes). Premium monitors use organic LED (OLED) technology, where each pixel is its own light source, offering superior contrast and color fidelity.

HDMI and DisplayPort Video Cables

Both HDMI (High-Definition Multimedia Interface) and DisplayPort cables have their distinct advantages, and it's important to select the appropriate cable based on the specific needs of the device and the desired performance outcomes.

High-Definition Multimedia Interface (HDMI)

HDMI cables are essential for multimedia setups due to its simplicity and versatility:

- **Functionality:** Transmits both high-definition video and audio signals through a single cable.
- **Standard Use:** Commonly used for connecting consumer electronics such as:
 - Televisions
 - Monitors
 - Gaming consoles
- **Versions:** Available in various versions, with the latest offering:
 - Higher resolutions
 - Increased refresh rates

- Advanced features like HDR (High Dynamic Range) for enhanced color and contrast
- **Advantages:**
 - Single-cable solution simplifies connections
 - Broad compatibility with a wide range of devices

DisplayPort

DisplayPort offers superior performance for high-resolution and multi-display configurations, making it a top choice for demanding visual applications.

- **Purpose:** Digital display interface for connecting a video source to a display device, such as a computer monitor.
- **Preferred Use:**
 - Professional environments
 - Gaming setups
- **Key Features:**
 - Supports higher resolutions and refresh rates than HDMI
 - Multi-Stream Transport (MST) allows multiple displays to connect through a single port, ideal for multi-monitor setups
- **Additional Capabilities:**
 - Transmits audio
 - Supports adaptive sync technologies like:
 - AMD FreeSync
 - NVIDIA G-Sync
 - Reduces screen tearing in gaming applications

Thunderbolt Interface

Although the Thunderbolt and Lightning interfaces are most closely associated with Apple computers and mobile devices, Thunderbolt is increasingly

implemented on Windows and Linux PCs too.

Thunderbolt can be used as a display interface like DisplayPort or HDMI and as a general peripheral interface like USB.

- Thunderbolt 1 and 2: Use the same physical interface as MiniDP and are compatible with DisplayPort, allowing a monitor with a DisplayPort port to connect via a Thunderbolt port and adapter cable. Thunderbolt ports are marked with a lightning bolt icon. Thunderbolt 2 supports up to 20 Gbps and allows daisy-chaining multiple monitors.
- Thunderbolt 3: Switches to the USB-C physical interface, using the same port, connector, and cabling. Converter cables are available for connecting Thunderbolt 1 or 2 devices to Thunderbolt 3 ports. USB devices function normally in Thunderbolt 3 ports, but Thunderbolt devices won't work in non-Thunderbolt USB-C ports. Thunderbolt 3 supports up to 40 Gbps over short, high-quality cables (up to 0.5 meters).
- Thunderbolt 4: Maintains the USB-C interface and supports up to 40 Gbps, with improved minimum performance requirements and expanded capabilities, such as support for docks with up to four Thunderbolt 4 ports.
- Thunderbolt 5: Also uses the USB-C interface, offering even higher speeds and enhanced features, though specific details may vary as the technology evolves. Thunderbolt ports are versatile, supporting high-speed data transfer, video output, and daisy-chaining multiple devices.

The USB-C form factor adopted for Thunderbolt 4

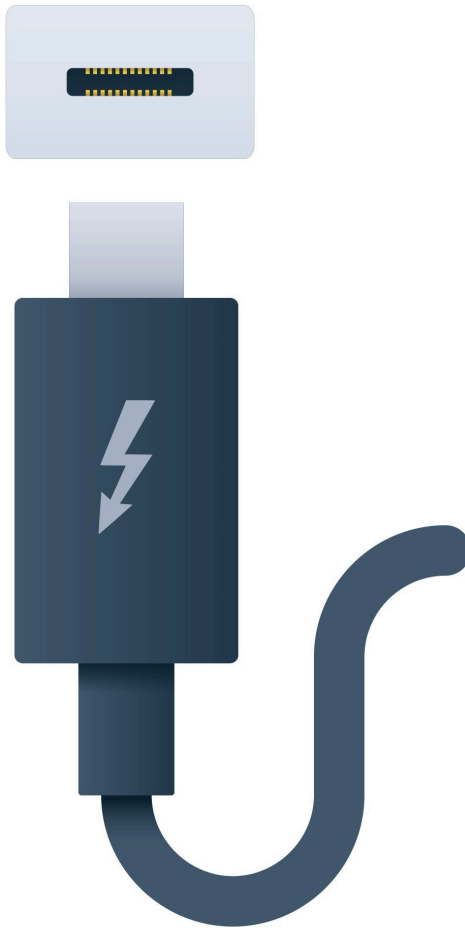


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Note:

Not all USB-C ports support Thunderbolt interfaces. Look for the flash icon on the port or check the system documentation to confirm.

Lightning Interface

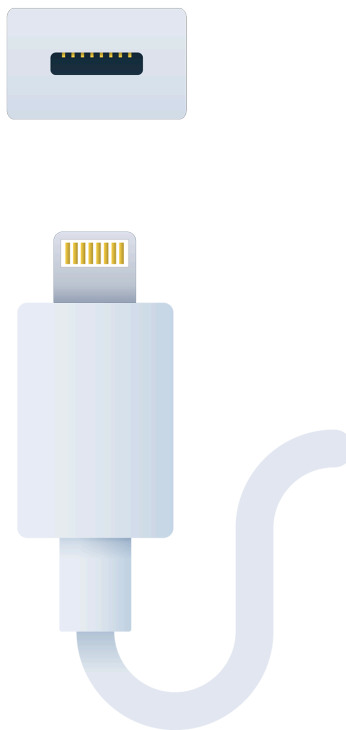
The Lightning interface, developed by Apple, is a proprietary connector used primarily in iPhone and iPad devices. Introduced in 2012, it replaced the older 30-pin dock connector with a more compact and reversible design, enhancing user convenience.

Exclusively found in Apple's mobile devices, the Lightning port requires a Lightning-to-USB A or Lightning-to-USB C adapter cable for connections to PCs and other devices, facilitating both charging and data transfer.

However, the Lightning interface is being phased out in favor of the USB-C standard. This transition is driven by:

- **Industry Standardization:** USB-C is widely adopted for its versatility, faster data transfer, and higher power delivery.
- **Regulatory Pressure:** Bodies like the European Union advocate for a common charging standard to reduce electronic waste and improve convenience.
- **Technological Advancements:** USB-C offers superior features, making it a more future-proof option.

Apple Lightning connector and port



Serial Advanced Technology Attachment Interface

As well as external cabling for peripheral devices, some types of internal components use cabling to attach to a motherboard port.

Serial Advanced Technology Attachment (SATA) is the standard for connecting internal storage drives in desktop PCs, using up to 1-meter cables with compact 7-pin connectors, each supporting a single device.

SATA connectors and ports (from left to right): SATA data, SATA power (with 3.3V orange wire)

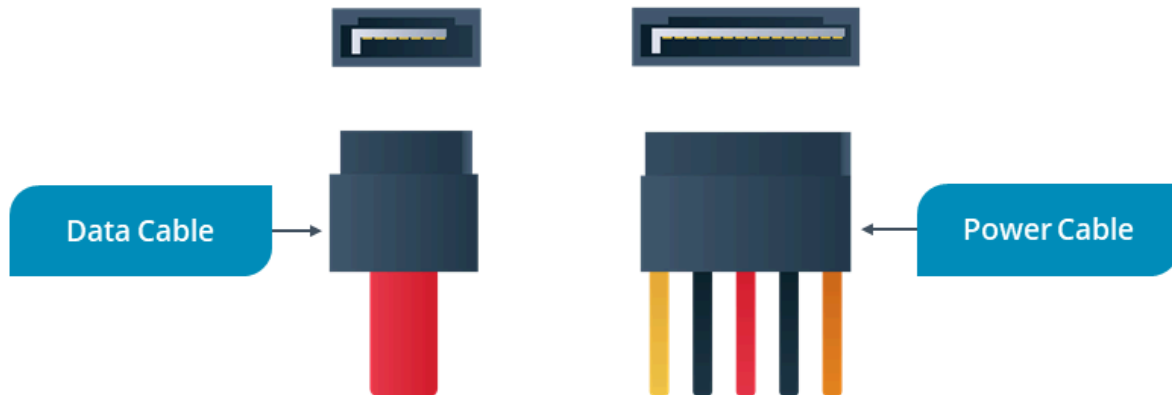


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The 7-pin data connector does not supply power; a separate 15-pin connector is used for power. The initial SATA standard supported speeds up to 150 MBps, which were later increased to 300 MBps with SATA revision 2 and 600 MBps with SATA revision 3.

SATA Revisions 3.1 to 3.5 introduced enhancements like the SATA Universal Storage Module, SATA Express specification, and better integration with I/O protocols but did not increase speeds.

Motherboard SATA and legacy PATA/IDE ports

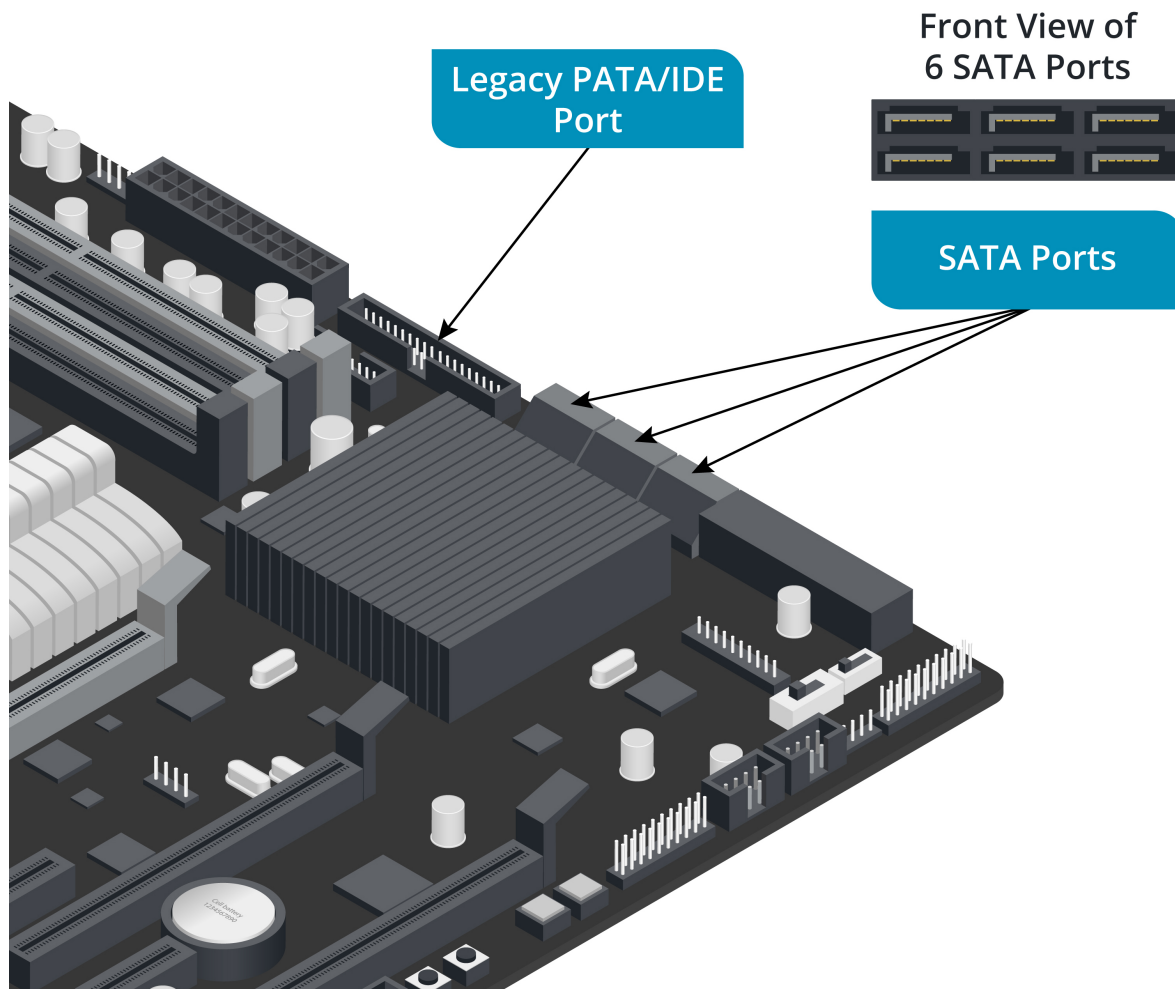


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Molex Power Connectors

Internal storage device data cables are designed solely for data transfer and do not carry power. To supply power, newer devices utilize a SATA power connector, whereas legacy components connect to the power supply unit (PSU) using a 4-pin Molex connector. These connectors are typically made of white or clear plastic.

The Molex connector features wire insulation color codes to indicate different voltage levels: red for 5 volt direct current (VDC), yellow for 12 volt direct current (VDC), and black for ground. Direct current (DC) refers to the unidirectional flow of electric charge, which is essential for powering electronic components.

Molex connector



Image ©123RF.com

Note:

Some devices might have both SATA and Molex power connectors.

External SATA

The External Serial Advanced Technology Attachment (eSATA) standard allows peripheral drives to connect using a cable up to 2 meters (78 in.) in length. An

eSATA cable is required for connection to an external eSATA port; internal SATA cables are not compatible. Some vendors offer a nonstandard powered port called eSATAp, which works with both USB and SATA using an eSATAp cable. Despite this, the USB interface remains the dominant choice for external drives.

Legacy Cables

DVI and VGA Video Cables

HDMI and DisplayPort interfaces are designed for digital flat-panel displays but can be adapted for analog devices. Older interfaces like VGA were used for cathode ray tube (CRT) monitors and projectors, which relied on analog signals.

Note:

DVI and VGA support only video, but not audio.

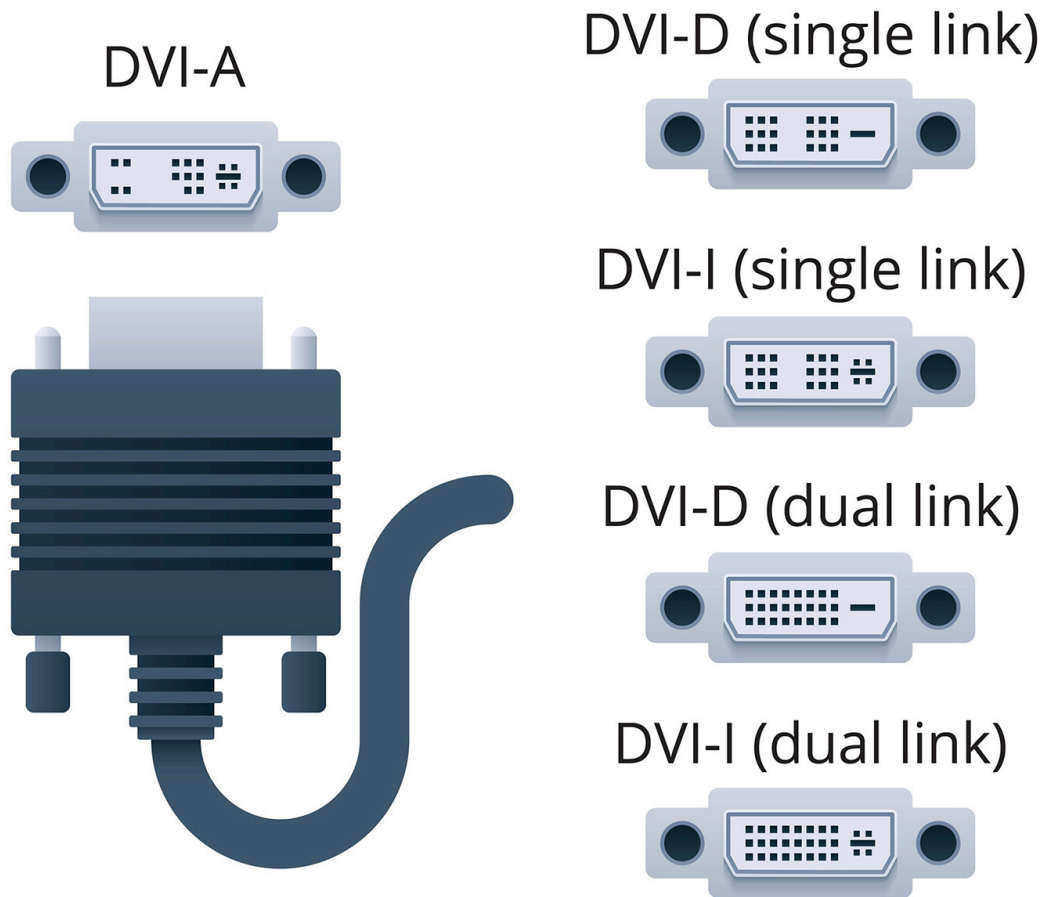
Digital Visual Interface

Digital video interface (DVI) supports both analog and digital outputs. Although popular after its 1999 introduction, DVI is no longer actively developed and is typically found on older display devices and video cards.

There are five types of DVI, each supporting different configurations for single and dual link (extra bandwidth) and analog/digital signaling. The connector's pin configuration indicates the type of DVI supported by a port.

DVI port and connector types

Image ©123RF.com



DVI-I supports both analog and digital outputs. DVI-A supports only analog output, while DVI-D supports only digital output.

Video Graphics Array Interface

The 15-pin **video graphics array** (VGA) port was the standard analog video interface for PCs for many years. Until recently, most video cards and monitors included a VGA port, though it is now being phased out. VGA typically supports resolutions up to Full HD (1920×1080), depending on cable quality. The connector is a D-shell type with screws for secure attachment.

A VGA connector and port



Image ©123RF.com

Serial Cables

The **serial** port is a legacy connection interface that transmits data one bit at a time over a single wire. Start, stop, and parity bits format and verify data transmission. This interface is also known as Recommended Standard #232 (RS-232). While modern interfaces like USB are also serial, RS-232 uses less sophisticated signaling, supporting data rates up to about 115 Kbps.

9-pin serial connector and port

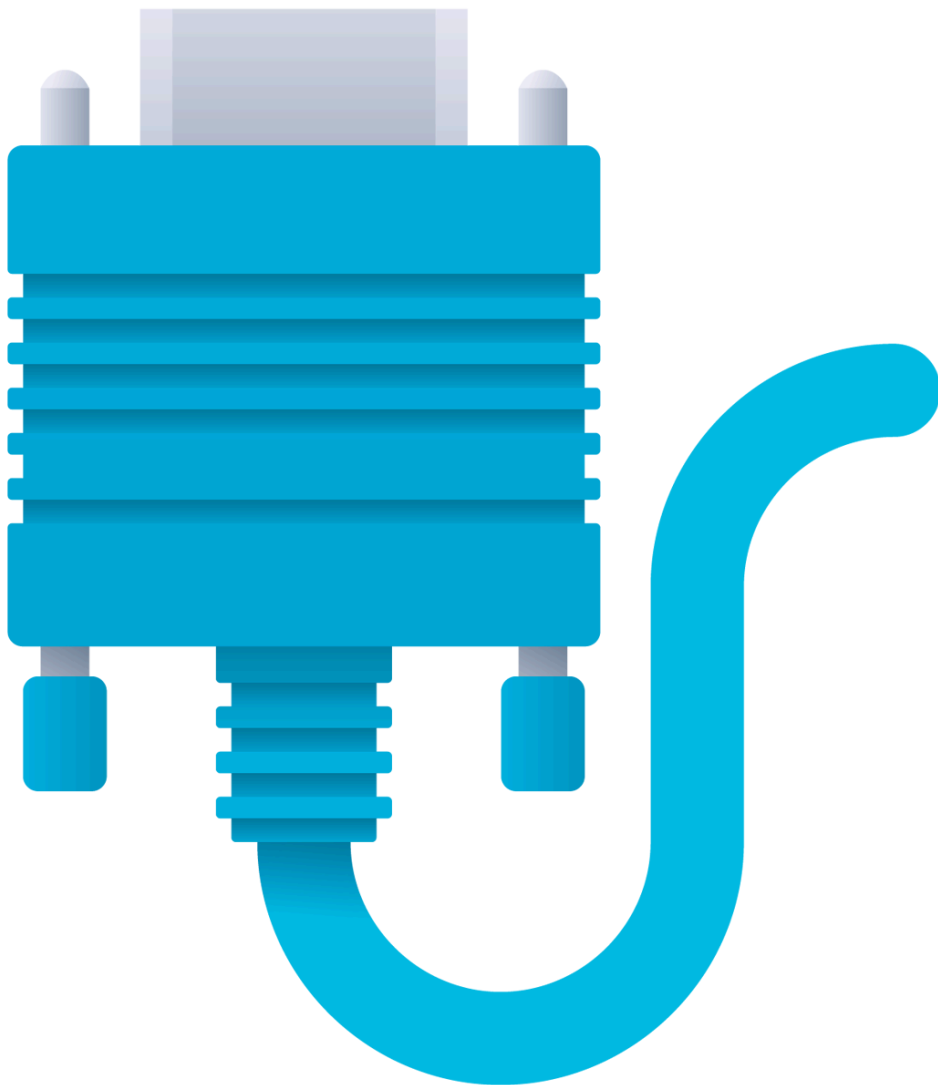
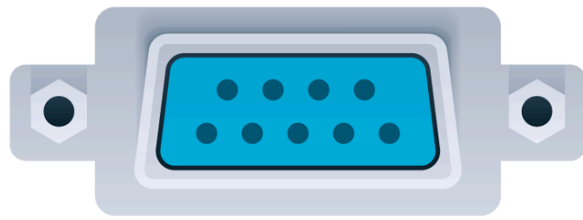


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Serial ports were commonly used to connect external modems for dial-up Internet, though USB has largely replaced this function. You may also find serial ports on network equipment for device management.

RS-232 specifies a 25-pin interface, but PC manufacturers typically use the cheaper 9-pin D-subminiature (**d-subminiature shell connector**) female port, commonly referred to as DB9, as shown above.

In Windows, the serial port is referred to as a Communications (COM) port.

Note:

You might also encounter PS/2 ports, used for connecting mice and keyboards. PS/2 ports use a 6-pin mini-DIN format, with green for mice and purple for keyboards.

Adapter Cables

Given the numerous cable and connector types available, a basic peripheral cable often won't directly connect a PC port to a peripheral device port. An adapter cable can solve this issue. An **adapter cable** has connectors for two different cable types at each end and comes in two types: **active adapters**, which use circuitry to convert signals (e.g., digital to analog), and **passive adapters**, which simply convert between connector types without changing the signal. Active adapters are necessary for converting signals that require different protocols, like digital HDMI to analog VGA. Passive adapters work when both connectors use the same signaling standard, such as different USB connector types.

Common types of adapter cables include:

- **Video adapters:** Convert between signaling types, such as **HDMI to VGA**, **HDMI to DisplayPort**, or **HDMI to DVI**.
- **USB adapters:** Convert between USB connector types, such as **USB-C to USB-A**.
- **USB hubs:** Provide additional USB ports, allowing multiple devices to connect to a single USB port.

- **USB adapters to various outputs:** Include adapters like **USB-C to HDMI** or **USB-C to Lightning**.